Field2VC: Automating the Generation of ASPICE-Compliant Verification Criteria from Field Failures

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***Abstract*—In Automotive SPICE, every SYS.2 system requirement must be verifiable through explicit verification criteria (VCs); for safety-relevant ECUs, ISO 26262 reinforces this obligation. Yet writing VCs that anticipate real-world field failures is slow and systematically incomplete: engineers rarely foresee failure modes they have not previously encountered. We present Field2VC, which derives VCs from two complementary failure records—developer-reported defects on GitHub, which expose code-level boundary conditions absent from specifications, and consumer-reported complaints in the NHTSA ODI database, which expose vehicle-level failures engineers do not anticipate. Field2VC extracts each report into a structured failure representation, retrieves the failures most relevant to a given requirement, and prompts a large language model to convert this evidence into specific, testable VCs. We contribute (i) an industrial benchmark of 350 de-identified SYS.2 requirements with expert-written reference VCs, and (ii) an empirical study in which four automotive domain experts assessed 489 generated VCs. Grounding generation in field failures raises expert acceptance from a 45.3 % no-context baseline to 65.5 %–67.2 % (odds ratio 2.3–2.5×; Holm-corrected $p < 0.001$; Cohen's $h = 0.41$–0.45, a small-to-medium effect), consistently across body-control, communication-stack, and diagnostics requirements. These are majority-vote labels with split decisions resolved by a coordinating adjudicator; §VI reports their robustness under a stricter no-adjudication assumption.**

***Index Terms*—verification criteria, automotive requirements, ASPICE SYS.2, issue mining, LLM, failure knowledge, ISO 26262, NHTSA**

I. Introduction

In automotive software engineering, the cost of repairing a defect grows exponentially across the development lifecycle. Boehm and Basili's empirical survey finds that fixing a defect after delivery can cost up to 100 times more than fixing it during the requirements phase [1] (Fig. 1). In addition, ASPICE SYS.2 strictly requires every system requirement to be *verifiable*: a *Verification Criterion* (VC) must coexist with the requirement text [2]. For safety-relevant ECUs this verifiability obligation is reinforced by ISO 26262, whose functional-safety lifecycle requires each requirement to be verified before integration [3].

Writing thorough VCs, however, fails in a specific and repeatable way. A requirement specifies intended behaviour under nominal conditions; a complete VC must additionally probe boundary conditions and fault modes. Reviewers reason forward from the requirement text, so the criteria they write can only confirm behaviours the specification already anticipates, while field failures, almost by definition, occur where it did not. The blind spot is structural rather than a matter of effort: a VC that omits an unforeseen edge case passes internal review precisely because no reviewer holds the evidence that would expose it (Fig. 1).

Closing this gap requires failure evidence from outside the project, matched to the abstraction level at which a SYS.2 requirement is verified—the *system* level, not the software unit. Defect trackers are a natural candidate, mined by recent work such as CrUISE-AC [4] to enrich agile acceptance criteria: developer-reported defects describe faults as engineers observe them—isolated to a module, reproducible, often with a root cause. But failures emerging from cross-subsystem interactions and months of customer usage rarely surface in any repository, because no developer observes the deployed fleet. Regulator-collected complaints capture exactly this stratum: NHTSA's ODI database records faults as the vehicle exhibits them, at fleet scale, under conditions no test plan enumerates [5], [6]. The two corpora are thus not interchangeable; our evaluation confirms the split empirically—complaint-grounded VCs are accepted more often for body and chassis requirements, defect-grounded VCs for communication and diagnostics (§V-B). We use NHTSA as a public proxy for the warranty and field-return data an OEM holds internally; the pipeline is agnostic to the vehicle-level channel's source. Manual inspection of 200 sampled requirement–evidence pairs confirms this mapping stage at **73.5 %** precision (§IV).

This motivates our central question: given a system requirement, can generation grounded in cross-level field-failure evidence produce verification criteria that practising engineers would accept into a SYS.2 baseline? We treat expert acceptance as the measurable proxy for usefulness, since no oracle for "complete" VC coverage exists.

This paper presents **Field2VC**, an automated VC generation mechanism grounded in real failure evidence. Field2VC extracts defect context from two channels—GitHub (code-level failures) and NHTSA (vehicle-level failures)— filters domain-specific noise from the raw records, and uses the retained context as a semantic constraint so a large language model (LLM) generates verifiable VCs covering historical failure scenarios.

a) Contributions.:

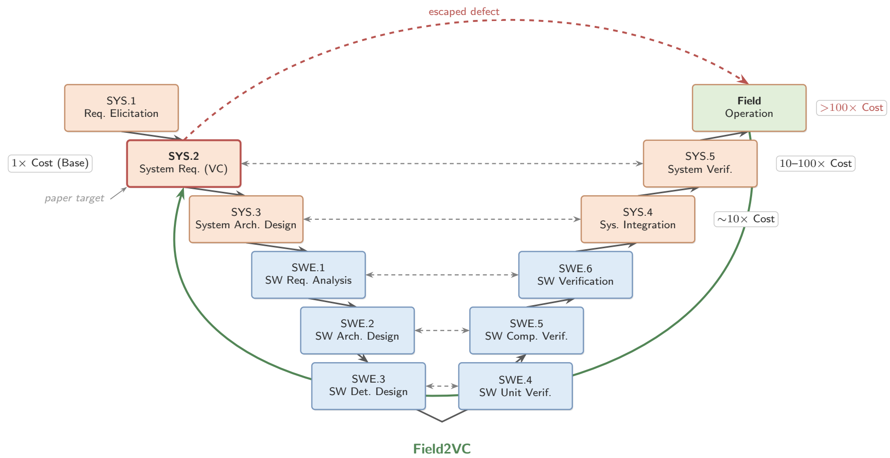

Fig. 1. ASPICE V-model cost escalation. A missing VC that passes SYS.2 review surfaces only at system-level testing or as a field complaint, where re-verification cost is 10–100× or more [1]. Field2VC shifts detection left by grounding VC generation in field-failure evidence *at* SYS.2 (highlighted by the *paper target* annotation); the green arc shows the evidence feedback loop from field operation back to SYS.2. SWE-layer processes (SWE.1–SWE.6) are outside Field2VC's scope: they receive the generated SYS.2 VCs as upstream constraints and are not modified by the pipeline.

- **An automated, evidence-grounded pipeline** combining developer-reported defects (GitHub) and regulator-collected complaints (NHTSA) as complementary channels at distinct abstraction levels, with an abstention mechanism that declines when retrieved evidence is insufficient (§III).
- **A four-expert evaluation** in which four domain experts (one an author) assessed 489 generated VCs spanning 307 unique industrial requirements, showing dual-channel evidence raises acceptance significantly above a no-context baseline (§V).
- **A quantitative characterisation of the two channels**, which cover complementary subdomains and collectively enhance verification coverage (§V-B).
- **An industrial benchmark**: a de-identified dataset of 350 real automotive system requirements (307 entering expert evaluation) with expert-annotated golden VCs, all generated VCs, per-rater labels, and analysis scripts (Data Availability).

Our results extend work on issue-tracker-driven requirements enrichment [4] in two directions that prior work does not cover: the evidence is drawn from two channels at *distinct abstraction levels*, and the target artefact is an ASPICE-mandated verification criterion for a safety-relevant ECU rather than an agile acceptance criterion.

b) Data Availability.: The de-identified replication package is available *now* for review at https://anonymous.4open.science/r/apsec2026-artifact-C562/, containing: (1) the SYRS corpus (all 350 benchmark items) with expert-annotated golden VCs; (2) all 489 generated VCs with per-VC rater labels, majority-vote labels, and the coordinating expert's adjudication record; (3) the Framework B evaluation guidelines; (4) pipeline scripts for Stages 2–5 and all prompt templates; and (5) statistical analysis scripts. Proprietary signal names are masked by stable placeholders; the redaction script is part of the package.

II. Related Work

A. Automated Requirements Completion within Specifications

Requirements completeness research falls into two streams. *Refinement* approaches improve individual requirements through semantic enrichment without adding new external data, using techniques such as NL-level templating and domain-knowledge-driven pattern systems, e.g., the EARS controlled requirements syntax [7]. *Derivation* approaches generate new requirements by exploiting relationships within or between existing requirement items.

LLM-based derivation has advanced rapidly [8]. F2SRD [9] retrieves the applicable OWASP ASVS verification requirements for each functional specification and guides GPT-4 to derive security requirements grounded in them. Such approaches draw their completion signal from within-specification logic or applicable standards, not from empirical failure evidence; our work is complementary: we exploit *external* failure repositories to surface conditions that no single specification can reveal.

Closest in spirit is TVR [10], which applies retrieval-augmented generation (RAG) to validate and recover missing traceability links between stakeholder and system requirements in industrial automotive projects. TVR targets *traceability* (linking existing requirements) rather than *VC completion* (generating new complementary criteria); its retrieval corpus is the specification itself, whereas ours is external failure evidence. These two directions show that RAG-based requirements engineering is being applied across multiple automotive sub-tasks [11], [12].

B. Enriching Requirements from External Repositories

Mining bug reports and issue trackers for RE has a long history [13], [14]. Do et al. [15], [16] extract requirements from similar products via doc2vec and BIRCH clustering; Almonte et al. [17] prompt LLMs to derive quality-attribute NFRs from functional specifications; Koscinski et al. [18] use RelGAN to transform functional information into security requirements. These mine requirements-like data from whatever corpora are available; we instead mine two repositories from *different domains*—a defect tracker and a regulator-operated complaint database—and show the knowledge source strongly predicts VC quality.

CrUISE-AC [4] is the leading crowd-data-driven system over external repositories, mining public issue trackers (seven e-commerce systems, $>165\,\mathrm{k}$ records, plus two content-management systems) for issues matching Gherkin acceptance criteria, achieving 82 % approval in e-commerce under a four-expert ($\geq$3-of-4) majority vote with Gwet's AC1 = 0.74. Wang et al. [19] likewise generate acceptance criteria from multi-modal requirements with LLMs, but target user-facing rather than safety-critical requirements.

Field2VC differs from CrUISE-AC in three respects. First, our domain is the automotive system-requirements level (ASPICE SYS.2) for safety-relevant ECUs. Second, we evaluate against a three-criterion Framework B standard (Completeness ≥ 3, Novelty = Y, Usefulness = Y) rather than a binary judgement, making direct percentage comparisons misleading. Third, our ablation contrasts LLM classification and embedding matching as grounding mechanisms, primarily exposing the effect of the filtering-enabled abstention decision rather than the matching method alone (§V-C).

C. LLM-based Requirements Engineering

Recent work uses LLMs for diverse requirements tasks [20]–[23]: MARE [24] proposes a multi-agent collaboration framework for requirements elicitation. Like the others, it targets neither VC completion nor the effect of grounding in external failure evidence—the key contribution of Field2VC. In the automotive-safety domain specifically, LLMs have been applied to code generation and verification [25]–[27], functional-safety and security design [28], and RAG-based derivation of safety requirements from standards [29]; these operate on code or upstream safety requirements rather than completing verification criteria for existing SYS.2 specifications.

III. Field2VC: Two-Channel VC Completion Pipeline

Field2VC is a five-stage pipeline that transforms raw failure records from two complementary external channels into candidate missing Verification Criteria (VCs) for automotive ECU SYRS items, each VC traceable to one or more source records (Fig. 2). We illustrate it throughout with a real UDS Security Access requirement (SYRS.1018_B_B) and its GitHub field failure (`python-udsoncan#51`)—a Channel A trace that surfaces protocol-transport boundary conditions absent from specification. Table I complements it with a Channel B (NHTSA) trace, where complaint evidence surfaces a real-world electromechanical scenario the golden VC was not designed to anticipate.

TABLE I
Worked trace: a representative accepted Channel B (NHTSA) VC from the main evaluation, tracing the full five-stage chain. Failure text quoted verbatim from the public source; SYRS and golden VC paraphrased from the de-identified corpus. The complaint surfaces a system-level interaction scenario beyond the nominal-path golden VC.

Channel B: NHTSA, SYRS.0177 (low-beam robustness under AC flicker). *SYRS:* do *not* switch off low-beam headlights when exposed to synthetic AC-flicker light sources if ambient brightness remains $< 1000\,\mathrm{lux}$ for more than $4\,\mathrm{s}$. The golden VC tests the *nominal AC-flicker path only*.
Matched failure: ODI #11557628 (sim. 0.60; 2022 Toyota Camry): "low beam headlights flashing when high beams are on . . . also flash on ignition."
Triple: ⟨high-beam activation or ignition key-on | low-beam LEDs remain steadily illuminated | bilateral intermittent flashing (right front more severe); state-stuck fault⟩.
Generated VC: fault-injection in an AC-flicker environment ($<$1000 lux sustained $>$4 s): activate high beams and separately cycle the ignition, verifying that low beams neither flash nor extinguish in either case—electromechanical interaction scenarios absent from the golden VC. **Accepted** (E3: C=3; E4: C=3; N = Y, U = Y; unanimous).

A. Stage 1: Fetch

We define two channels providing complementary failure evidence:

Channel A: GitHub Issues. We index issues from four open-source repositories covering ECU-adjacent stacks: `pylessard/python-udsoncan` (UDS diagnostics), `python-can-isotp` (ISO-TP transport), `hardbyte/python-can` (CAN bus), and `ecubus/EcuBus-Pro` (ECU test toolchain). We retrieved **696 issues** via the GitHub REST API,[1] prioritising closed issues labelled {bug, defect, regression, safety} and supplementing with unlabelled closed issues for coverage. The running example fetches `python-udsoncan#51`: *"SecurityAccess does not handle already-unlocked server."*

Channel B: NHTSA Complaints. We query the NHTSA ODI consumer-complaint database [30] using component keywords matching automotive ECU subsystems. We retrieved **364 complaint records**, each containing a free-text consumer narrative plus structured vehicle and component metadata. Unlike curated recall notices, complaint narratives are raw, colloquial, and noisy field reports, which is why the Stage-2 filter matters most for this channel.

B. Stage 2: Filter

We use an LLM (Qwen-Plus) to filter each raw record into a structured triple:

$$\langle \textit{Trigger} \mid \textit{Protocol Behavior} \mid \textit{Fault Mode} \rangle$$

[1] https://docs.github.com/en/rest/issues

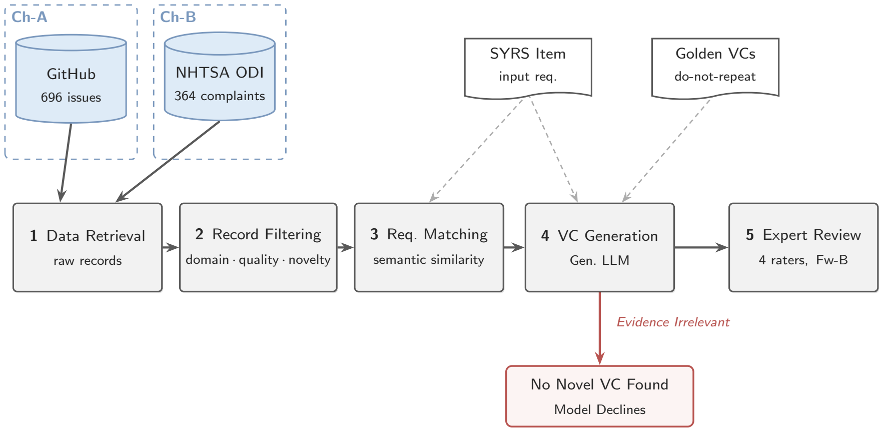

Fig. 2. Field2VC five-stage pipeline. Failure records from GitHub (Ch-A) and NHTSA (Ch-B) are filtered into ⟨Trigger | Protocol Behavior | Fault Mode⟩ triples (Stage 2, Qwen-Plus), embedding-matched to each SYRS item via cosine similarity (Stage 3, `text-embedding-v3`), and used to prompt VC generation (Stage 4, Qwen-Max). The generator is never forced to produce a VC: it may decline by emitting the fixed marker `NO_NOVEL_VC_FOUND` when the evidence reveals no genuinely missing condition, the pipeline's only relevance gate. Four domain experts review the surviving VCs under Framework B (Stage 5).

Trigger is the activating condition, *Protocol Behavior* the expected ECU action per the relevant standard, and *Fault Mode* the observed deviation; filtering reduces vocabulary mismatch and suppresses implementation-layer detail irrelevant to SYS.2. The core instruction is *"Extract the technical essence from this report"*, with a mandatory escape route: records not about automotive system behaviour (e.g., Python packaging, documentation) return all three fields as `null`[2] and are dropped—this is where off-topic noise (packaging, documentation, consumer venting) leaves the pipeline. The running example, issue #51, reduces to ⟨`0x27` requestSeed while already unlocked | client detects all-zero seed and skips sendKey | key derived from zero seed → NRC `0x24` rejection⟩.

C. Stage 3: Match

For each SYRS item s and filtered triple t_i we compute cosine similarity [31]–[33] using `text-embedding-v3` (Alibaba DashScope; snapshot April–May 2026), retrieving the top-$K = 5$ triples as the *failure context* $\mathcal{F}(s) = \{t_1, \ldots, t_5\}$. We impose *no* hard similarity cut-off: every item proceeds with its top-5 context and the relevance decision is delegated to generation, where the model abstains (`NO_NOVEL_VC_FOUND`) if the evidence reveals no genuinely missing condition (§III-D). This mirrors, post-retrieval, the pre-retrieval LLM relevance classifier of CrUISE-AC [4]—a choice our ablation examines directly (RQ3, §V-C). For SYRS.1018_B_B, issue #51 ranks second at cosine 0.69 and enters the top-5 context. Embedding matching has two advantages over LLM binary classification: it scales to thousands of pairs without per-pair LLM calls, and it yields a continuous, auditable relevance score released with every match.

[2]All Stage-2 and Stage-4 prompts are released verbatim in the replication package (`prompts/`).

D. Stage 4: Generate

We prompt Qwen-Max with three inputs: the SYRS item, its existing golden VCs (as *do-not-repeat* context operationalising "missing"), and the failure context $\mathcal{F}(s)$. The prompt requires VCs that (a) genuinely test the requirement (not the protocol library the failure was reported against), (b) are not redundant with the golden VCs, and (c) are inspired by a real failure pattern in $\mathcal{F}(s)$ and testable at ECU system-integration level. It closes with explicit permission to decline—*"if no genuinely useful missing VC can be identified, output* `NO_NOVEL_VC_FOUND`*"* (the *sentinel*). Declining is expected, not an error: rather than hallucinate a plausible criterion [34], the pipeline abstains when the evidence is irrelevant. For the running example, the model turns issue #51's triple into a negative test of the already-unlocked path ($27 L1 unlocked, send `$31 01 10 08`, expect no NRC $33)—a scenario neither the SYRS nor its golden VC exercises.

E. Stage 5: Review

Each generated VC is independently scored by at least two of the three external raters (E2, E3, E4) on three dimensions under **Framework B**:

- **C (Completeness)**: Score 1–5. Does the SYRS item omit this VC? ($C \geq 3$ is the acceptance threshold.)
- **N (Novelty)**: Y/N. Is this VC absent from the existing SYRS item and its associated VCs?
- **U (Usefulness)**: Y/N. Is this VC actionable for downstream test design and software qualification?

A VC is accepted if and only if $C \geq 3$ AND $N = \text{Y}$ AND $U = \text{Y}$. The label is the majority vote among the raters who scored each VC; E1 (adjudicator) resolves disagreements on the N and U dimensions. The running-example VC was accepted (C=4, N=Y, U=Y) after E1 adjudicated a novelty split between E2 and E3.

F. Ablation Variants

Two ablation conditions isolate the contributions of filtering and embedding matching. **Abl-NF (No Filter)** skips Stage 2, embedding the raw issue title + body (first 500 characters) and matching to SYRS by cosine similarity—testing whether raw-text embedding suffices. **Abl-LM (LLM-Match)** replaces Stage 3 similarity with an LLM binary relevance classifier (Qwen-Plus), replicating CrUISE-AC's matching [4] to test LLM judgement against embedding similarity.

IV. Experiment Design

A. Research Questions

- **RQ1 (Effectiveness):** Does failure-grounded prompting generate VCs with significantly higher expert acceptance than a no-context baseline?
- **RQ2 (Channel–Domain Fit):** The two channels are mapped to disjoint requirement subdomains by construction. Within each subdomain (body-control/chassis, communication-stack, diagnostics), how effective is the assigned channel relative to the no-context baseline, and how do the two channels divide labour across subdomains?
- **RQ3 (Ablation):** What is the contribution of (a) failure-triple filtering and (b) embedding-based matching to VC quality?
- **RQ4 (LLM Sensitivity):** How stable are the channel-level results across different LLMs from multiple model families?

B. Dataset

a) SYRS corpus.: We apply the pipeline to 350 unique automotive ECU SYRS items drawn from a proprietary corpus of 675, covering body control (ESCL, EPB, SCCM), communication stack (UDS, CAN), diagnostics, and system-state management. The channel mapping is disjoint by construction: Ch-A covers 208 communication-stack and diagnostics items, Ch-B 142 body-control, HMI, and functional-safety items. All items follow the ASPICE SYS.2 format (explicit actor, trigger, quantitative boundary, expected behaviour) and carry human-annotated golden VCs.

b) Generated VCs.: Across both channels and the no-context baseline, 489 VCs entered expert evaluation (Table II):

TABLE II
DATASET AND PER-CONDITION VC FLOW. CH-A AND CH-B COVER DISJOINT SYRS SUBSETS ($208 + 142 = 350$) AND ABSTAIN VIA THE NO_NOVEL_VC_FOUND SENTINEL; "CAND." COUNTS CANDIDATE VCS AND THE CAND. → EVAL. GAP REMOVES DUPLICATE/MALFORMED CANDIDATES. SOURCES: 696 GITHUB ISSUES, 364 NHTSA COMPLAINTS.

Condition	Items	Abst.	Cand.	Eval.
A (GitHub)	208	121	88	87
B (NHTSA)	142	13	131	128
Baseline (all SYRS)	350	73	277	274

Evaluation totals $87 + 128 + 274 = 489$ VCs spanning 307 unique SYRS; 43 items yielded no evaluated VC. Ch-A and Ch-B share no SYRS; the baseline spans all 350 items (overlapping both channels).

C. Evaluation Metrics

a) Accept rate.: Fraction of VCs whose majority-vote label is Accept (Framework B: $C \geq 3$ AND $N = \text{Y}$ AND $U = \text{Y}$).

b) Wilson 95 % CI.: Confidence interval for the accept rate using the Wilson score method [35].

c) Fisher's exact test + Holm–Bonferroni correction.: Each channel is tested against the baseline with Fisher's exact test [36]; p-values are corrected with Holm–Bonferroni [37] across three simultaneous comparisons (Ch-A, Ch-B, and A+B combined vs. baseline).

d) Cohen's h.: Effect size for proportion comparisons [38]: $h \geq 0.2$ small; $h \geq 0.5$ medium; $h \geq 0.8$ large.

e) Generation-rate proxy.: The *generation rate* is the fraction of SYRS items for which the model generates a VC rather than abstaining (`NO_NOVEL_VC_FOUND`). Under active filtering a high rate indicates genuinely new failure angles, while near-universal generation (>93 %) signals the model is ignoring the abstention instruction. We use it as a lightweight quality proxy for the RQ4 sensitivity runs, where full human evaluation is unavailable.

f) Abstention rates.: Ch-A abstains on 121 of 208 SYRS items (58.2 %), Ch-B on 13 of 142 (9.2 %). The gap reflects vocabulary alignment: once filtered, NHTSA fault modes map tightly to body-control and chassis items, while GitHub vocabulary is more varied relative to communication-stack and diagnostics items. The asymmetry is itself a finding: the pipeline generates selectively, not indiscriminately.

g) Matching precision.: To establish confidence in the upstream retrieval quality, we manually evaluated a stratified random sample of 200 requirement–evidence pairs (100 per channel, seed = 42, similarity range 0.45–0.78). A pair is labelled *relevant* if the retrieved evidence describes a failure mode or boundary condition observable in the ECU under test (not merely a topically adjacent software defect or host-toolchain issue). The pipeline achieves an overall matching precision of **73.5 %** (Ch-A GitHub: **61.0 %**; Ch-B NHTSA: **86.0 %**), confirming that the evidence reaching the generator is failure-relevant.

D. Inter-Annotator Agreement

a) Rater profiles.: Four raters span the three ASPICE V&V stakeholder roles (process ownership, academic methodology, hands-on practice), guarding against single-perspective bias: **E1** (Embedded Systems Project Manager, 10 yr) adjudicates split votes under defined criteria; **E2** (ASPICE assessor and functional-safety consultant) contributes SYS.2 process authority; **E3** (PhD candidate in automotive software testing) provides methodology-grounded judgement; **E4** (senior functional-test engineer, OEM/Tier-1) anchors physical V&V executability. This industry–academia panel replicates CrUISE-AC's design [4].

b) Main evaluation (RQ1/RQ2).: E2, E3, and E4 independently scored the 489 VCs (each receiving two independent external ratings); E1 resolved split decisions under Framework B, and the label is the majority vote among scoring raters. We use Gwet's AC1 [39], [40] rather than Cohen's κ to avoid the kappa paradox [41] under high Accept-class prevalence, reported on Completeness ($C \geq 3$).

c) Ablation evaluation (RQ3).: The 50-item subset comprises the first 50 Ch-A items in corpus order, all Communication-Stack requirements; this homogeneity strengthens internal comparability (category cannot confound the contrast) but restricts RQ3's external scope to that category (§VI). All four raters (E1–E4) score the subset; the label is the majority vote of E2–E4 (≥2/3) with E1 as adjudicator (matching the main protocol), and all four scores feed the pairwise AC1 analysis.

E. Baselines

a) No-context baseline.: Qwen-Max prompted with SYRS text only (no failure context). This represents the zero-shot VC generation scenario and is the primary comparison point for RQ1.

b) CrUISE-AC [4].: The closest published system, targeting acceptance criteria for user stories from crowd-reported bugs. Accept rates and IAA values are compared in §V-A to contextualise our results.

F. Computational Cost

The core pipeline (Qwen-Plus filtering, Qwen-Max generation on both channels, and the ablations) cost approximately **US$3** in API fees; the seven-model RQ4 sweep added roughly US$12 (dominated by reasoning-heavy models' long completions), for a total on the order of US$15. Embedding matching was negligible (< $0.05) and the Domain-2 pilot added about $0.5. Figures are approximate, depending on time-varying provider pricing.

V. Results and Analysis

A. RQ1: Effectiveness of Failure-Grounded Prompting

Table III summarises acceptance rates for all conditions. Channels A and B both significantly outperform the baseline after Holm–Bonferroni correction. Channel A (GitHub) achieves 65.5 % acceptance (57/87), OR = 2.30, Cohen's $h = 0.41$, Holm-$p = 0.0007$. Channel B (NHTSA) achieves 67.2 % (86/128), OR = 2.48, $h = 0.45$, Holm-$p < 0.001$. The combined A+B pipeline reaches 66.5 % (143/215), confirming both channels contribute; the joint A+B vs. baseline Fisher test yields $p < 0.001$, OR = 2.40, $h = 0.43$.

TABLE III
RQ1: Accept Rates vs. Baseline (three-criterion protocol; majority vote)

Condition	VCs	Accept	Rate	Holm-p
Baseline (no-context)	274	124	45.3 %	—
Channel A (GitHub)	87	57	65.5 %	0.0007***
Channel B (NHTSA)	128	86	67.2 %	<0.001***
A+B combined	215	143	66.5 %	<0.001***

*** $p < 0.001$ (Holm–Bonferroni corrected).

a) IAA.: Pairwise Gwet's AC1 on Completeness ($C \geq 3$) is 0.84 (E2–E3), 0.65 (E2–E4), and 0.69 (E3–E4), averaging 0.73; the two lower pairs both involve E4's physical-executability perspective vs. E2/E3's requirements/process perspectives (a discipline-level finding, §VI). The mean is comparable to CrUISE-AC's AC1 = 0.74 [4], under our strictly stronger three-criterion acceptance predicate.

b) Per-item accepted-VC yield.: Conditional acceptance rates compare *generated* VCs and are not commensurable across conditions with different abstention rates. A common-denominator metric is the *per-item yield*: accepted VCs over all items assigned to a condition. Generation rate, accept rate, and yield are 79.1 %/45.3 %/0.35 for the baseline (350 items), 41.8 %/65.5 %/0.27 for Ch-A (208 items), and 90.8 %/67.2 %/0.61 for Ch-B (142 items).

Ch-A's yield (0.27) is below the baseline (0.35): it generates for only 41.8 % of items, though 65.5 % of those are accepted—a *precision channel* rather than a high-recall supplement, best combined with the baseline where broad coverage is needed. Ch-B's yield (0.61) substantially exceeds the baseline: complaint-grounded VCs cover most assigned items and are accepted at a higher rate.

B. RQ2: Channel–Domain Fit

Because the two channels are mapped to disjoint requirement subdomains, RQ2 concerns where each channel fits rather than which channel performs better in a head-to-head comparison. Within its assigned domain, each channel yields a category-dependent uplift over the no-context baseline averaging +20 pp (Channel A +20.7 pp, Channel B +21.3 pp; category-matched,[3] derived from Table IV), each supported by an independent small-to-medium effect size relative to the baseline (Channel A $h = 0.41$; Channel B $h = 0.45$). The grounding mechanism differs by source: once Stage-2 filtering removes their colloquial noise, NHTSA complaints' component-tagged fault modes align closely with SYRS vocabulary, whereas GitHub issues offer higher volume (696 vs. 364 records) at the cost of greater terminological variance.

a) Scope-stratified acceptance.: Table IV breaks acceptance down by ECU category. Body-control and chassis items (Ch-B) show consistently high acceptance—IO/HMI 74.2 % (+28 pp), Functional Safety 71.9 %, Body Control 68.3 %—as filtered NHTSA narratives map tightly to sensor-level and state-machine specifications; communication-stack items (Ch-A) improve more modestly (61.3 % vs. 56.4 %, +4.9 pp), reflecting greater GitHub-issue vocabulary variation for protocol-specific SYRS items.

Table I traces an accepted Channel B VC end-to-end; with the Channel A running example of §III it illustrates the two channels' complementary discovery roles.

C. RQ3: Ablation

Table V reports human acceptance rates on the 50-item Ch-A evaluation subset under the three ablation conditions.

[3] *Pooled* deltas (the +20/+22 pp in RQ1) compare each channel against the full baseline; *category-matched* deltas (RQ2) compare against the baseline restricted to that channel's categories.

TABLE IV
SCOPE-STRATIFIED ACCEPTANCE BY ECU CATEGORY (THREE-CRITERION PROTOCOL, MAJORITY VOTE). "—" = NOT IN BASELINE CORPUS.

Category	Ch.	Pipeline		Baseline	
		Acc/N	Rate	Acc/N	Rate
Body-Control & Chassis (Ch-B)					
IO / HMI	B	23/31	74.2 %	18/39	46.2 %
Functional Safety	B	23/32	71.9 %	—	—
Body Control	B	28/41	68.3 %	21/42	50.0 %
Comm. Stack & State Mgmt (Ch-A/B)					
Communication Stack	A	19/31	61.3 %	31/55	56.4 %
System State Mgmt	B	12/24	50.0 %	11/28	39.3 %
Diagnostics (Ch-A)					
Diagnostics (DCM)	A	29/44	65.9 %	25/64	39.1 %
Diagnostics (DEM)	A	4/6	66.7 %	14/33	42.4 %
Diagnostic Services	A	5/6	83.3 %	4/13	30.8 %

TABLE V
RQ3: HUMAN ACCEPTANCE RATE ON 50-ITEM CH-A SUBSET (4-RATER MAJORITY VOTE, C $\geq$ 3 AND N=Y AND U=Y).

Condition	Accepted	Rate
Full Field2VC (FL, filter + embed)	29/50	58.0 %
Abl-LM (LLM-Match*)	29/50	58.0 %
Abl-NF (No Filter, embed)	41/50	82.0 %

*Replicates CrUISE-AC matching strategy [4]: LLM binary relevance classification, no triple filtering.

a) Embedding matching vs. LLM binary classification.: On the same 50-item subset, the clean matching-method comparison is between the two *unfiltered* ablations: Abl-NF (raw-text embedding) achieves 82.0 % versus 58.0 % for Abl-LM (LLM binary classifier, equivalent to CrUISE-AC's strategy [4]), a 24 pp gap attributable to the retrieval mechanism since filtering is absent in both. The full pipeline (FL, filter + embed) matches Abl-LM exactly at 58.0 %: with filtering active, the accept rate is set by the Stage-2 abstention decision, not the matching step.

b) Why filtering is necessary at scale.: We do not dispute the local inversion: removing the filter (Abl-NF, 82.0 %) scores higher than the full pipeline (58.0 %) on these 50 items. Its cost is invisible at this scale and surfaces only at corpus scale, where Abl-NF's generation-rate proxy reaches 96–100 %: without the ⟨Trigger|Protocol Behavior|Fault Mode⟩ triple, raw embedding matches thematically misaligned issues and the generator produces a VC for nearly every SYRS regardless of relevance. For Ch-A the pipeline generates for 87 of 208 items (41.8 %), accepting 65.5 % of those—it abstains when evidence does not support a grounded VC. The full pipeline thus trades bounded subset-level precision (58.0 % vs. 82.0 %) for the suppression of near-universal ungrounded generation, the trade-off required to move from a 50-item demonstration to a corpus-scale process. The proxy is meaningful only when filtering is active (§V-D).

c) IAA (FL condition).: On the 50-item FL evaluation, individual accept rates were E1 33/50 (66.0 %), E2 24/50 (48.0 %), E3 47/50 (94.0 %), E4 27/50 (54.0 %), for a majority vote (E2+E3+E4 $\geq$2/3) of 29/50 (58.0 %). Six rows produced E1–MV disputes—five where E1 accepted but MV rejected (ExtAddr scope mismatch or tool-layer contamination), one (Row 013) where E1 rejected but MV accepted (BufferedReader ECU wake-up boundary)—with MV prevailing per protocol. Mean 4-rater Gwet's AC1 is 0.63 for Abl-NF (substantial) and 0.37 for Abl-LM (moderate), corroborating the quality difference; pairwise agreement spans 66–92 % (Abl-NF) and 34–88 % (Abl-LM), the E3–E4 low driven by methodological differences on Extended-Addressing conditions (§VI).

D. RQ4: LLM Sensitivity

TABLE VI
RQ4: LLM SENSITIVITY. THE PRIMARY RUN IS HUMAN-EVALUATED (ACCEPT RATE); SENSITIVITY RUNS REPORT THE GENERATION-RATE PROXY ONLY (NOT COMPARABLE TO THE ACCEPT RATE). RUNS ARE INDEPENDENT RE-RUNS, SO QWEN-MAX'S CH-A RATE HERE (40.4 %) DIFFERS FROM THE MAIN RUN (41.8 %) BY NORMAL VARIANCE.

LLM	Ch-A	Ch-B
Primary run — human accept rate		
Qwen-Max	65.5 %	67.2 %
Sensitivity runs — generation-rate proxy		
Sentinel-adherent		
Qwen-Max (sensitivity)	40.4 %	93.7 %
Qwen-Plus	59.6 %	94.4 %
Non-sentinel-adherent (>93 % on both channels)		
DeepSeek-v4-Pro	99.0 %	100.0 %
DeepSeek-v4-Flash	100.0 %	99.3 %
Kimi-K2.6	94.2 %	97.2 %
GLM-5	98.6 %	98.6 %
Qwen3.6-Plus	100.0 %	99.3 %

a) Sentinel adherence divides model families.: Five of seven models generate for >93 % of items regardless of knowledge-source relevance, effectively ignoring the `NO_NOVEL_VC_FOUND` sentinel. Only Qwen-Max and Qwen-Plus emit it discriminatively: Qwen-Max (sensitivity) yields 40.4 % for Channel A (most varied GitHub vocabulary), rising to 93.7 % for Channel B; Qwen-Plus shows a shallower gradient (59.6 %, 94.4 %). The proxy is thus a reliable quality signal only when the LLM respects the suppression instruction. Across both sentinel-adherent (Qwen-family) models the B > A ordering holds, consistent with RQ2's scope-stratified analysis.

VI. DISCUSSION AND THREATS TO VALIDITY

A. Failure Mode Analysis

Analysis of the 72 rejected VCs across Channels A and B reveals three failure modes that account for the channel acceptance gap.

FM-1: Boundary Underspecification (the most common mode). The VC is well-formed but lacks quantitative parameters (timing constraints, byte constants, thresholds), because source issues describe qualitative fault behaviour without numeric bounds and the LLM omits boundaries that appear only in the SYRS text. More common in Channel A; Channel B's component metadata and filtered fault modes constrain the generated bounds. *Recommendation:* inject all SYRS numeric literals as mandatory prompt constraints, and flag VCs that reference none.

FM-2: Scenario Transplant (dominant Channel-B mode). The VC identifies the right ECU component and fault type but embeds the test in an irrelevant consumer-driving scenario lifted from an NHTSA narrative (e.g., "drive… 30 minutes with slight steering inputs"), making it impossible on a standard HIL bench. *Recommendation:* a scenario-abstraction step that strips road-context sentences and substitutes laboratory-equivalent stimuli.

FM-3: Filtering Collapse (thematic misalignment at scale). Without Stage 2 filtering, embedding similarity alone retrieves topically adjacent but thematically misaligned issues, and the LLM generates coherent VCs testing conditions unrelated to the SYRS under test (e.g., a UDS service-0x29 test for a framing requirement). This mode surfaces *at corpus scale*, not on the 50-item subset: across the full corpus Abl-NF generates for 96–100 % of items (§V-C), whereas the filtering-enabled pipeline abstains on the majority (58.2 % of Ch-A SYRS, generating for only 41.8 %)—the gap is the volume of misaligned matches that filtering suppresses. *Recommendation:* filtering is not optional for safety-critical applications; at minimum, enforce an AUTOSAR module-tag guard before matching.

B. Threats to Validity

a) Construct validity.: The accept/reject label depends on raters applying Framework B ($C \geq 3$ AND $N =$ Y AND $U =$ Y). To mitigate subjectivity we recruited four raters—three independent scorers (E2–E4) and an adjudicator (E1)—with automotive RE backgrounds, and computed Gwet's AC1 [39] to account for the kappa paradox [41] under high Accept-class prevalence. E1 was not blind to the generation channel and is an author, a confirmation-bias risk—particularly for Channel B (67.2 %), which relied heavily on E1's adjudication. We anchored E1's rulings to the pre-registered Framework B and cross-verified all 58 disputed Channel-B acceptances against the rulings of E2 (an independent ASPICE assessor). Beyond these safeguards we report the headline rates under two alternative adjudication assumptions rather than claim independence (E1 resolved 50/87 Channel-A, 89/128 Channel-B, and 146/274 baseline splits). Under a conservative *no-E1 floor* (every split a reject), Channel A falls to 42.5 % against a 43.4 % baseline and Channel B to 15.6 %: the per-channel uplift does not survive this worst case, so the headline effect is partly adjudication-dependent. Restricted to the *agreed subset* (items both external raters labelled identically), Channel A is accepted on 37/37 items (100 %) versus 119/128 (93.0 %) for the baseline—so the Channel-A effect persists without adjudication on clear-cut cases—while Channel B accepts only 51.3 %, confirming it leans more on adjudication than Channel A. Single-adjudicator, non-blind scoring thus remains a genuine limitation: unlike CrUISE-AC [4], which accepts on ≥3-of-4 *independent* experts, our protocol pairs two external raters with an adjudicator, and a larger blinded, independent panel under majority vote is the natural strengthening. The E2–E4 and E3–E4 Completeness pairs (0.65, 0.69) are the lowest and both involve E4, reflecting genuine disciplinary divergence between E4's physical-executability perspective and the requirements/process perspectives of E2 and E3, not a measurement failure. The generation-rate proxy (RQ4) approximates novelty without human evaluation but cannot rank novel VCs by quality; its validity is confirmed only when filtering is active (§V-C).

b) Evaluator methodology divergence (Abl-LM).: The Abl-LM condition shows markedly low E3–E4 agreement (34 %, AC1 = −0.24), a systematic methodological difference, not random noise. E3 applied a *protocol-mechanism* criterion grounded in ISO 15765-2 (since N_Bs/N_Cs timers and STmin decouple from addressing mode, an Extended-Addressing (ExtAddr) precondition is treated as non-modifying context and judged on protocol-level merit), whereas E2 and E4 applied a *deployment-readiness* criterion (an ExtAddr condition inconsistent with a physical-addressing bench renders the VC non-deployable). Both reflect legitimate practice; majority vote (≥2/3) lets the conservative E2+E4 position prevail, making the 58.0 % Abl-LM rate a conservative lower bound.

c) Internal validity.: The Fetch stage uses repositories chosen for ECU-adjacent vocabulary; we mitigate by drawing from three distinct stacks (UDS diagnostics, CAN transport, ECU toolchain). A small prompt-budget asymmetry remains: the do-not-repeat golden-VC context is truncated at 800 characters in the pipeline conditions but 600 in the baseline; we disclose rather than re-run, as it affects only that context, not the failure evidence. The absence of a hard similarity gate (§III-C) is supported post-hoc: every VC surviving sentinel abstention had a top-1 similarity ≥0.50, and above this floor similarity does not predict acceptance (quartile accept rates 57–76 %, non-monotonic); a hypothetical 0.60 gate would have discarded VCs with a *higher* accept rate (70.0 %) than those retained (63.5 %). The Channel-B trace (Table I) makes this concrete: at 0.60 top-1 the retrieved complaint reports an *internal electrical* perturbation (low-beam flashing on high-beam/ignition) rather than the *external optical* AC-flicker stimulus SYRS.0177 specifies, yet Stage 4—conditioned on the SYRS and its golden VC, not the evidence alone—recovered the shared concern (low-beam ON-state stability) into a unanimously accepted VC. Gate-free retrieval supplies coarse candidates while requirement-grounded generation keeps the VC anchored to SYRS intent even when the evidence mechanism diverges—a scope condition, not a guarantee, expected only where the requirement states its intent precisely enough to anchor generation, as SYRS.0177's bounds (<1000 lux, >4 s) do.

d) Evidence grounding vs. abstention.: One might ask whether the gains reflect the failure evidence or merely the Stage-2 abstention mechanism inflating the accept rate by reporting only a favourable subset. We separate the two using per-item yield (accepted VCs over *all* assigned items), which charges every abstention as a miss. Channel B attains its $+21.3\,\text{pp}$ category-matched uplift while abstaining on only $9.2\,\%$ of items $(129/142$ generated), so it cannot be attributed to selective silence; Channel A instead abstains on $58.2\,\%$ and its yield (0.27) falls below the baseline (0.35), trading recall for precision. We therefore claim that failure grounding improves VC quality where filtered evidence aligns with requirement vocabulary (Channel B), not that it uniformly raises generation; an abstention-matched head-to-head isolating the marginal effect of evidence is left to future work.

e) DUT-boundary filtering failure (RQ3).: In the 50-item FL evaluation, 14 of the 21 MV-Reject VCs (67 %) arose from DUT-boundary leakage: python-can host-layer properties—log levels, file-descriptor limits, Windows socket IDs, periodic-sender timing bugs (hardbyte/python-can#888)—were extracted as ECU-observable behaviours and passed the triple filter, because issue threads mix host- and device-side concerns and Stage 2 cannot infer which side of the DUT boundary a property belongs to without a component-layer annotation. The resulting VCs ask the ECU to verify conditions only the test harness can observe. For SYRS.0693_A, e.g., the generator asserted correct behaviour at file-descriptor counts beyond 1023 and at specific host log levels (`python-can` properties, not ECU-observable), which all but one rater rejected—a discriminated rejection confirming the evaluation is not rubber-stamping plausible output. A guard checking for known test-tool namespaces (`python-can`, `udsoncan`, `isotp`) before triple extraction would suppress most such false positives.

f) External validity.: The primary evaluation covers one proprietary project (Channel A: communication-stack and diagnostics; Channel B: body-control, HMI, and functional-safety). To probe generalisability we applied Channel A (Qwen-Max) to a separate Domain-2 pilot of 85 Diagnostics SYRS items (DCM, DEM, Diagnostic Services), which produced 15 novel VCs (17.6 % generation-rate proxy; mean top-1 similarity 0.625). A full three-rater evaluation (E2–E4) under Framework B accepted **13 of 15** (86.7 %, Wilson 95 % CI 62–96 %), exceeding the primary Channel-A rate (65.5 %); agreement was substantial for Novelty (mean AC1 = 0.69) and moderate for Usefulness (0.42, the latter from E4's stricter physical-executability criterion, e.g. rejecting DoIP VCs on CAN-only benches, with E2–E3 at AC1 = 1.00). The lower generation rate vs. Ch-B (17.6 % vs. 90.8 %, 129/142) reflects more self-contained Diagnostics items; the overlapping CI gives only preliminary evidence that cross-domain transfer succeeds without domain-specific prompting. Further limits: the B > A ordering holds only on two Qwen-family models, so cross-family stability is untested; powertrain, ADAS, and gateway domains remain unexamined, and NHTSA effectiveness may not transfer where no regulator-operated complaint database exists; and the RQ3 ablation subset is homogeneous (all 50 items Communication-Stack), so its conclusions do not automatically extend to other categories.

g) Conclusion validity.: We apply Holm–Bonferroni correction [37] across three Fisher's exact tests (family-wise $\alpha = 0.05$), report Wilson 95 % CIs [35] for all rates, and report Cohen's h [38] alongside p-values to curb over-interpretation in small-to-medium samples. Several scope-stratified cells (Table IV) are small (Diagnostics DEM and Diagnostic Services each n=6), with wide Wilson intervals; we report them as descriptive breakdowns, not powered comparisons.

VII. Conclusion

We presented Field2VC, a five-stage pipeline that completes automotive ECU verification criteria by grounding LLM generation in empirical failure evidence from two complementary external channels: GitHub issues (Ch-A) and NHTSA consumer complaints (Ch-B). A four-expert evaluation (three independent scorers and one author-adjudicator) of 489 generated VCs spanning 307 unique ASPICE SYS.2 items yields four findings:

RQ1. Failure-grounded prompting with Ch-A and Ch-B achieves 65.5 %–67.2 % expert acceptance versus a 45.3 % no-context baseline (Holm-corrected $p < 0.001$, $h = 0.41$–0.45, IAA: mean Gwet's AC1 = 0.73).

RQ2. Within their assigned subdomains both sources are effective (category-matched uplift: Channel A +20.7 pp, $h = 0.41$; Channel B +21.3 pp, $h = 0.45$). Once filtered into structured fault modes, NHTSA's component-tagged descriptions align closely with SYS.2 vocabulary, giving the strongest gains in body-control and chassis categories (68–74 %) and more modest improvement in communication-stack items (61.3 %).

RQ3. On the 50-item ablation subset, FL and Abl-LM both accept $58.0\,\%$ (29/50); the clean embedding-vs-LLM-match contrast is Abl-NF (82.0 %) vs. Abl-LM (58.0 %), a 24 pp gap attributable to matching since filtering is absent in both. When filtering is active, acceptance is driven by the Stage-2 abstention decision, not retrieval. Filtering is necessary at scale: without it, near-universal generation (96–100 %) indicates ungrounded matching.

RQ4. The B > A quality ordering holds across both sentinel-adherent (Qwen-family) models—a knowledge-source-driven, not LLM-driven, pattern—while five of seven models fail to respect the generation-suppression sentinel (>93 % generation rate), making sentinel adherence a prerequisite for using the generation rate as a quality proxy.

A Domain 2 pilot on 85 Diagnostics SYRS (UDS/DTC) offers preliminary cross-domain evidence: of 15 novel Channel-A VCs under full three-rater evaluation (E2/E3/E4), 13 (86.7 %) were accepted, exceeding the primary Channel-A rate (65.5 %) without domain-specific prompt engineering.

Two reusable lessons emerge: (1) *a failure source's usefulness depends on whether, after filtering, its vocabulary aligns with the quantitative, component-level language safety-critical requirements demand*—regulator complaint records satisfy this more consistently than community issue reports; and (2) *generation-rate proxy validity depends on the LLM's adherence to generation-refusal instructions*, which must be validated before use.

To support replication, we release the full de-identified dataset—the 350-item SYRS corpus (307 evaluated) with golden VCs, all 489 generated VCs with per-rater labels and adjudication, the Framework B protocol, the LLM prompts, and the analysis scripts—as a reproducible benchmark (Data Availability).

a) Future Work.: Planned extensions include automated VC formalisation into machine-checkable predicates, a traceability matrix linking each accepted VC to its source failure record, and replication on powertrain and ADAS domains.

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